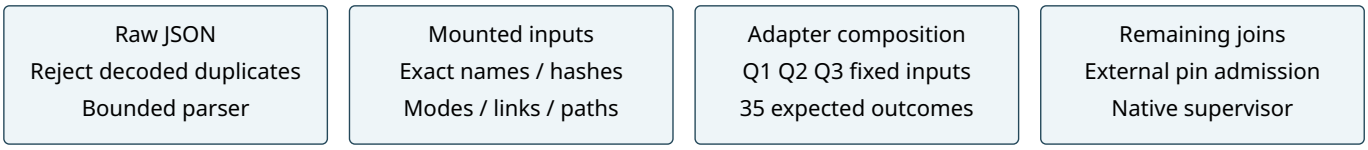
Native JSON and mounted-input custody composition

The Rust input layer rejects duplicate decoded JSON keys recursively, trailing values, malformed/nonfinite JSON and byte-cap violations. It composes exact mounted-fixture inventory, file and gate digests, canonical path spelling, symlink rejection, Unix type/mode/link checks and descriptor/path stability observations with the previously reviewed metadata-and-diff adapter. Thirty-five fixed software controls passed their expected outcomes. No gate binary, daemon, collector or scientific job was executed.



Source connection. The frozen supervisor's strict parser, verify_fixture_members, gate checks and prepared/fixtures snapshot supply the source predicates. The imported adapter is byte-identical to SHA256 91a8f422b708307bf305baca65f661ab3ddf4a578d6d663f19cc54c9bec0f116. The new qualify_inputs entry point parses raw inspect and expected-identity bytes, joins both mount sources to caller-supplied custody pins, checks custody before and after the adapter, and rejects changed snapshots. Fixed Q1, Q2 and Q3 metadata inputs reach this complete entry point.

Boundary controls. Rejections include nested and escaped duplicate keys, invalid UTF8, excess JSON/file bytes, wrong digests, empty or extra fixture inventories, parent and relative paths, symlinked directories, hard-linked files, writable modes, mismatched mount-source identity, failed diff transport and a duplicate raw inspect field. Fixture files contain inert bytes; all mutations are confined to a new private fixed-control directory.

Remaining interfaces. Custody pins remain supplied by a trusted caller; this module does not admit its own pins or authenticate a review. It covers the flat mounted prepared/fixtures directory and the standalone gate, not the outer preparation manifest, full runtime archive, build provenance or all admission joins. File descriptor and metadata checks do not prove absence of transient races or bind a later daemon mount. The native supervisor lifecycle, process/daemon observations, cleanup and deadlines remain separate. File size caps are caller inputs, not changes to campaign limits. The parser follows serde_json numeric/depth limits rather than promising arbitrary-precision Python JSON equivalence.

References checked. The original frozen supervisor and prior adapter review were read first. RFC 8259 section 4 documents the interoperability problem with duplicate object names; section 9 permits parser size/depth/number limits (https://www.rfc-editor.org/rfc/rfc8259). Rust canonicalize documentation explains that it resolves symlinks and normalizes components (https://doc.rust-lang.org/std/fs/fn.canonicalize.html); this module additionally requires exact canonical spelling and checks symlink metadata. No new mathematical or runtime-performance claim is made.

Verification status. Offline Rust compilation and fixed controls returned exit zero. Producer source and exact dependency lock are retained for an independent source-fidelity review after publication. Historical Python supervisor code was read as source only; it was not invoked. The full 1,260-target research scope and expired scientific deadline remain unchanged.